

Parametric Faithfulness: CoT vs. ToT

Question

Does multi-path Tree-of-Thought reasoning expose a stronger dependence on the model's internal parameters than a single Chain-of-Thought path?

Core idea

- ▶ Generate reasoning traces for multiple-choice science QA.
- ▶ Unlearn selected reasoning steps while preserving held-out behavior.
- ▶ Re-evaluate whether the original answer flips after the intervention.

Runs completed

Dataset	Forget	Retain
OpenBookQA	50	20
ARC-Challenge	100	20

Model and update

- ▶ Llama-3.2-3B-Instruct
- ▶ NPO-KL unlearning
- ▶ FF2-only updates, 5 epochs
- ▶ same hyperparameters for CoT and ToT

Experimental Setup

Reasoning arms

Arm	Construction
CoT	Single visible reasoning path.
ToT-selected	Five sampled paths; choose the highest-confidence path.
ToT validation	sample-select vs. beam-prune before full runs.

Selected ToT mode

Dataset	Mode	Val. Acc.	Paths
OpenBookQA	sample-select	80.0	5.00
ARC-Challenge	sample-select	80.0	5.00

Metrics

- ▶ **Eff**: removal of the unlearned reasoning behavior.
- ▶ **Spec**: preservation on retained examples.
- ▶ **FF-HARD**: answer flips after unlearning among eligible questions.
- ▶ **FF-SOFT**: probability shift away from the original answer.
- ▶ **Post-CoT Agree**: post-reasoning answer agreement filter.

Integrity check

All four main runs completed with full question coverage. No malformed JSON, duplicate steps, missing epochs, tracebacks, or CUDA OOMs were found.

Main Results

Dataset	Arm	Eff	Spec	FF-HARD
OpenBookQA	CoT	81.2	95.4	72.5
OpenBookQA	ToT-selected	81.8	95.1	71.8
ARC-Challenge	CoT	75.8	95.6	68.8
ARC-Challenge	ToT-selected	79.6	95.4	72.2

Dataset	Δ Eff	Δ FF-HARD	Δ FF-SOFT
OpenBookQA	+0.6	-0.7	+5.2
ARC-Challenge	+3.8	+3.4	+7.3

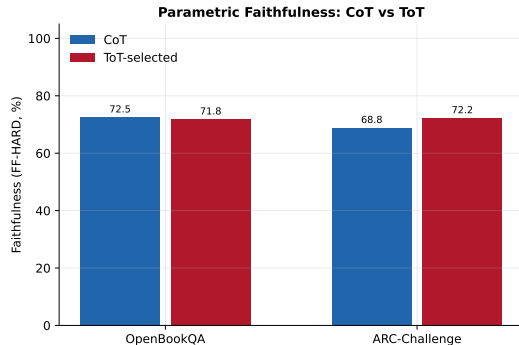
Takeaways

- ▶ ARC-Challenge shows the clearest ToT advantage: higher Eff, FF-HARD, and FF-SOFT with almost unchanged Spec.
- ▶ OpenBookQA improves in soft probability shift, but not in strict paper-protocol FF-HARD.
- ▶ Paired eligible subset: ARC gains +3.2 FF-HARD points; OpenBookQA is tied.

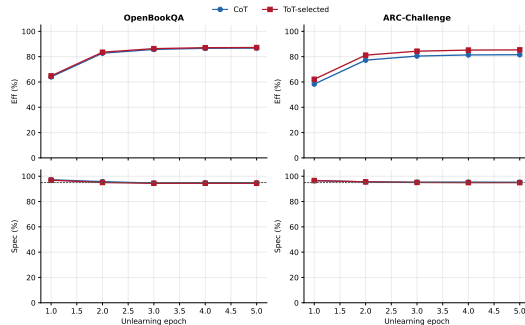
MMLU Gen was not run in this batch, so it is not imputed from other metrics.

Visual Summary

Faithfulness comparison



Unlearning trajectories



Bottom line: ToT-selected is most useful on ARC-Challenge, where multi-path reasoning exposes stronger parameter dependence while preserving retained behavior.